

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Botany

Questions: 20

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

Q2. Describe Algal Phylogeny. Describe sporophyte development in bryophytes with suitable /0 illustrations. /0 Discuss the molecular basis of plant-pathogen interaction, with examples. 10 Describe the role of bacteria in pharmaceuticals, agriculture and industry. /0

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2. [UNKNOWN] paper-i

Q2. (a) Compare loose smut with covered smut of wheat plants on the basis of symptoms, casual organism, disease cycle and disease management. 5+5+5=15 Describe the evolutionary significance of various stellar structures found in Pteridophytes. Give one suitable example in each case. 10+5=15 Describe the diversity of photosynthetic pigments found in the major classes of Algae. 10

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3. [UNKNOWN] paper-i

Q3. Distinguish between Simple and Cleavage Polyembryony. With the help of labelled diagrams, compare the embryogeny of Pinus and Cycas. 10 Enumerate the sporophytic and gametophytic features of Psilotum. Comment on its systematic position. /0 Discuss the role of transduction in genetic recombination in bacteria. /0 Characterize the different types of Mycorrhiza with examples. Discuss its role in agriculture and forestry. /0

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4. [UNKNOWN] paper-i

Q3. (a) What are the recent developments in the use of microbes in industrial products ? Why do microbes serve as most ideal organisms in industrial applications? 8+7=15 Explain the progressive sterilization of potentially sporogenous tissue found in Bryophyta. Add a note on the role of elaters. 5+5=10 Describe the structure and chemical composition of TMV. How does it infect the host? 10+5=15

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5. [UNKNOWN] paper-i

Q4. Define Telome. Discuss Zimmermann's telome concept citing examples from major pteridophytic groups. /0 Describe briefly the symptoms and control measures of black stem rust of wheat. Name the causal organism. Why is it known as macrocyclic rust? /0 Distinguish between Lytic and Lysogenic cycles. Explain briefly the /0 steps of multiplication process of T4-bacteriophage by lytic cycle. Discuss the role of amphibious plants in the evolution of land plants. /0 PSI-P-BTN 2 ===== PAGE 3 ===== SECTION B

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6. [UNKNOWN] paper-i

Q4. (a) Describe the organization of female cone in Pinus. Discuss on the morphological nature of ovuliferous scale. 10+5=15 Comment on how far Selaginella advances towards seed habit. 10 What is Systemic Acquired Resistance (SAR) ? Discuss the role of salicylic acid in SAR. 7+8=15 FSI-P-BTN 2 ===== PAGE 3 ===== SECTION B

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7. [UNKNOWN] paper-i

Q5. Answer the following keeping your answers brief and to the point: 8x5=40 Explain with examples, the different provisions of author citations. 8 Discuss Vavilov's Centres of diversity. 8 Give five botanical names of fibre yielding plants and mention the parts from where fibres are obtained. 8 Discuss in brief, the distinctive flora of the Mesozoic Era.

Give a concept on the Inflorescence of Musaceae. 8

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8. [UNKNOWN] paper-i

Q5. Answer the following keeping your answers brief and to the point. $8 \times 5 = 40$ How do you justify Asteraceae as a phylogenetically advanced family in dicotyledons? 8 Explain 'Nemec phenomenon' and its significance. 8 What are cortical vascular bundles? How are they formed? $3 + 5 = 8$ Discuss the role of Botanical Gardens in conservation of plants. 8 What is haploid cell culture? How is this technique useful in agriculture? 8

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9. [UNKNOWN] paper-i

Q6. (a) Discuss the basic outline of classification according to the principles of Bentham and Hooker. State its merits and demerits. $6 + 4 = 10$ Compare the floral characteristics of Fabaceae and Asclepiadaceae. Give floral diagrams and name an economically important plant of each of these families. $6 + 2 + 2 = 10$ What do you mean by Palynology? Give an account of application of palynology. $2 + 8 = 10$ Define Ethnobotany. Describe the role of ethnobotany in the discovery of modern drugs, citing examples. $1 + 9 = 10$

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10. [UNKNOWN] paper-i

Q6. (a) Give an account of Cronquist's classification of flowering plants. Add a note on merits and demerits of this system of classification. $10 + 5 = 15$ What is totipotency? Give an experimental evidence to demonstrate it. $5 + 10 = 15$ What is Helobial type of endosperm? Describe the process of its development. $4 + 6 = 10$

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11. [UNKNOWN] paper-i

Q7. (a) Write the botanical names, active constituents and uses of five plants of economic importance belonging to each of the families Fabaceae, Rosaceae and Brassicaceae. 15 Discuss the different levels of biodiversity. Comment briefly on in situ and ex situ conservation of biodiversity. State the role of cryopreservation in the conservation of plant biodiversity. $6 + 5 + 4 = 15$ What is Apomixis? Discuss different apomictic developmental pathways with suitable illustrations. Give a note on genetics of apomixis $1 + 5 + 4 = 10$ FSI-P-BTN 3 ===== PAGE 4 ===== QS. What is meant by Anomalous Secondary Growth? Give a descriptive account on comparison of normal and anomalous secondary growth in dicot with examples and illustrations, wherever necessary. $2 + 13 = 15$ Define Botanical Garden and state how it differs from other gardens. Name two internationally reputed botanical gardens. Discuss the role of an ideal botanical garden in education. $2 + 2 + 6 = 10$ Write notes on the following: $5 \times 3 = 15$ Micropropagation: Definition, stages and application 5 Callus Culture: Origin and application 5 Protoplast Fusion: Origin, merits and demerits 5 PSI-P-BTN 4

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12. [UNKNOWN] paper-i

Q7. (a) What is polyembryony? How is it induced? Discuss its applications. $5 + 5 + 5 = 15$ (b) Write the botanical name of the plants, their family, economically important plant parts and uses for the following: Cinnamon Clove Saffron Nutmeg Lesser Cardamom 15 (c) Compare the floral characters of Malvaceae and Solanaceae. Give the floral diagrams of the two families. $5 + 5 = 10$ F5I-P-8TN 3 ===== PAGE 4 =====

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13. [UNKNOWN] paper-i

Q8. (a) Name the four types of tea recognized in the trade. What are their properties and methods of processing? $2 + 6 + 7 = 15$ Describe the organization of essential organs of Asclepiadaceae family. Discuss the pollination mechanism in it $6 + 9 = 15$ Give an account on hydrocarbon plants and their potential as a source of sustainable energy. 10 FSI-P-STN 4

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14. [UNKNOWN] paper-ii

1. Write short notes on each of the following: $8 \times 5 = 40$ Complexes in electron transport chain of mitochondria. 8 Molecular

basis of sex determination. 8 Significance of accessory chromosomes. 8 Merits and demerits of GM plants in India. 8 1.(e). Chi-square test. 8 Describe the number and structural variations in chromosome. Add a note on their significance. 15+5=20 Describe with illustrations about chromosomal behaviour during meiosis with translocation, inversion and duplication. 10+10=20 Elucidate the role of RNA in the origin and evolution of living organisms. 10+10=20 With suitable example, explain the tripartite sex-determination. 20 1 fsi-d-btn ===== PAGE 2 =====

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15. [UNKNOWN] paper-ii

1. Write a short note on each of the following : 8x5=40 Packaging of DNA in chromatin Chloroplast dimorphism Post-transcriptional modifications of mRNA Role of endoplasmic reticulum in protein sorting and targeting Objectives and properties of probability

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16. [UNKNOWN] paper-ii

2. (a) Describe the molecular basis of cell cycle transitions with suitable illustrations. Explain the role of CDKs in controlling transitions between cell cycle stages. 10+10=20 (b) Discuss the mechanism of transport of molecules across the cell membrane by uniport, symport and antiport. Add a note on co-transport by symporter and uniporter. 15+5=20

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17. [UNKNOWN] paper-ii

3. (a) Describe the molecular mechanism of gene linkage and crossing-over. Explain how gene mapping is constructed using gene linkage with suitable diagrams. 10+10=20 (b) Describe different methods of selection and hybridization in plant breeding. Give a note on heterosis breeding. 15+5=20

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18. [UNKNOWN] paper-ii

4. (a) Describe different stages of micropropagation in detail, mentioning the significances of each stage. Give a brief account on commercial micropropagation. 15+5=20 (b) Describe the initiation of translation process in prokaryotes with suitable illustrations. How does it differ from eukaryotes? 15+5=20 SECTION—B

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19. [UNKNOWN] paper-ii

5. \$. (d) 5.(e) Explain the bio-chemical and molecular basis of mutations. 10+10=20 Narrate briefly the various methods of gene transfer in plants. 20 SECTION 'B' Write brief notes on each of the following: 8x5=40 Allelopathy 8 Intellectual property rights 8 Molecular markers in plant breeding 8 Role of carotenoids in plants 8 Evidences of organic evolution 8 "Calvin cycle is said to be autocatalytic" — Justify the statement. Compare the mode of CO₂ assimilation in C₃ and C₄ plants. How does C₄ pathway differ from C₃ pathway in terms of efficiency? 5+10+5=20 Describe the role of phytohormones in agri-horticulture. 20 With illustration describe the structure of phytochrome. Explain the role of phytochrome in flower development process. 8+12=20 Describe the methods of breaking seed dormancy. Trace the physiological changes involved during seed germination. 8+12=20 Phytoremediation is not an ultimate solution to all the maladies of air and water. Explain in detail with specific examples. 20 Describe the various phytogeographical zones met within India and add their salient features. In an ideal situation, there will be a high temperature and rainfall with restricted snowfall plus Rhododendron, Oaks, Orchids and Ferns — what is your inference? 15+5=20 fsi-d-btn 2

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20. [UNKNOWN] paper-ii

5. Write a brief note on each of the following : 8x5=40 Role of Rubisco in carbon metabolism Importance of different types of secondary metabolites in plant defense and as pharmaceuticals • • FSI-D-BTN/43 2 ===== PAGE 3 ===== Endemism and endangered plant species Hot spots in India : Characteristics and conservations (e) Allosteric enzymes and metabolic control (a) Illustrate the structure of ATP synthase and discuss the mechanism of ATP synthesis in higher plants. 5+15=20 (b) Explain elaborately the molecular basis of fruit ripening. How can this process be manipulated? 15+5=20 (a) Discuss the mechanism of nitrogen fixation associated with legumes and elaborate the factors controlling

the process. 10+10=20 (b) Describe the adaptive responses at the morphological and physiological levels in higher plants to water deficit stress conditions. 10+10=20 (a) Explain the ecological characteristics of different types of forests found in India. Describe briefly about the dominance of vegetation types of tropical and alpine forests. 10+10=20 (b) Discuss the concept of climax community. Write a note on succession of estuarine ecosystem. 10+10=20 * * *
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